7



PERSONAL GROWTH

4 PERSONAL GROWTH PROTOCOLS

1. Replace Mind Wandering with Present-Tethered Awareness
2. Apply James Hollis's Guide to Creating Your Best Life
3. Resolve Inner Pain Points Through Expressive Writing
4. Reframe Loss to Effectively Move Through Grief

Thus far I've focused on protocols, and the science they're based on, for bolstering key pillars of your health and performance: sleep, exercise, nutrition, stress management, light exposure, as well as focus, motivation, and learning. You'll notice I did so mostly through the lens of biology, not psychology. Even when talking about mindsets or ways to generate a sense of urgency, the emphasis unabashedly has been on the biological underpinnings of what makes these protocols work. That reflects my training and interests, but as I mentioned in the introduction it's also an essential part of how these protocols work. They are customizable to your needs. When you know how a machine works—how the parts fit together—you come away knowing which aspects you can modify.

We, however, are not machines. We think and feel, and hope and fear, and more. That needs attention too. The quality of our lives is strongly driven by our emotional lives, our relationships to others and to ourselves. After all, what good is it feeling vital and rested, focused and resilient, fit and capable, if we don't also enjoy our time with others, and with ourselves? We just can't overlook these aspects of life.

Scores of data now tell us the obvious: that social connection is vital to our health and longevity.[1] Like all the other Old

[1] Julianne Holt-Lunstad, "Social Connection as a Critical Factor for Mental and Physical Health: Evidence, Trends, Challenges, and Future Implications,"

World primates, we have an immense need to care for and be cared for by others.[2] Our nervous system evolved to crave and require these types of interactions we consciously or unconsciously register as safety.[3] Not just safety from harm, but also the sort of safety that makes us feel capable of seeking out challenges: healthy risk-taking in our work, new goals in our hobbies, going on adventures; all things that give us a sense that life, our life, matters.

Equally important but far less discussed is our relationship to ourselves. That's an admittedly squishy term. What I mean is how we talk to ourselves, and how we think about ourselves—the decisions we make, the paths we choose. In fact, this chapter could have been called "Protocols for Thinking." My goal here is not to get you to second-guess yourself. Quite the opposite. It's about clearing the clutter from your mind so you can think better. It's about moving forward on the right path and not mindlessly slaving away on the wrong one.

World Psychiatry 23, no. 3 (September 2024): 312–32, https://doi.org/10.1002/wps.21224.

[2] Anne E. Storey and Toni E. Ziegler, "Primate Paternal Care: Interactions Between Biology and Social Experience," Hormones and Behavior 77 (August 2015): 260–71, https://doi.org/10.1016/j.yhbeh.2015.07.024; Allan Schore, Right Brain Psychotherapy (W. W. Norton & Company): 2019.

[3] George M. Slavich, "Social Safety Theory: A Biologically Based Evolutionary Perspective on Life Stress, Health, and Behavior," Annual Review of Clinical Psychology 16 (March 2020), https://doi.org/10.1146/annurev-clinpsy-032816-045159.

What are thoughts anyway?

Marcus Aurelius famously said, “Our life is what our thoughts make it”[4] and nearly identical statements were made by René Descartes,[5] the Buddha,[6] Ralph Waldo Emerson,[7] William Shakespeare,[8] and in the Bible.[9] Most people would agree their thoughts are a crucial feature of their experience of life but few ever stop to ask, “What are thoughts?” Most neuroscience textbooks don’t even attempt to answer that question. They include chapters on sensation, perception, emotions, behaviors, even memories and dreams, but not a single textbook that I am aware of provides a good explanation of what thoughts are. Fortunately, researchers are starting to figure it out.

Jennifer Groh, PhD, neuroscientist at Duke University, researches how our brain maps, or represents, to use the neuroscience term, the world around us.[10] That makes her

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4 Marcus Aurelius, Meditations.

5 René Descartes, Discourse on the Method (1637) (“I think, therefore I am”).

6 Buddha, The Dhammapada, chap. 1, verses 1–2 (“What you think, you become”).

7 “Ralph Waldo Emerson: ‘A Man Is What He Thinks About All Day Long,’ ” The Socratic Method, September 21, 2023, https://www.socratic-method.com/quote-meanings/ralph-waldo-emerson-a-man-is-what-he-thinks-about-all-day-long.

8 William Shakespeare, Hamlet, act 2, scene 2 (“There is nothing either good or bad but thinking makes it so”).

9 Proverbs 4:23 (“Be careful how you think; your life is shaped by your thoughts”); Proverbs 23:7 (“For as a man thinks in his heart, so he is”).

10 Jennifer M. Groh, Making Space: How the Brain Knows Where Things Are (Belknap Press of Harvard University, 2014).

particularly well suited to answer the question “What are thoughts?” because thoughts are representations too. They are estimates, approximations, best guesses. She explained to me, “When we are thinking, we are running simulations with the sensory and motor infrastructure of our brain.” She gave me the following example, posed as a brief series of questions. If you want, you can answer them now as well.

Think of a cat.

What comes to mind first?

Chances are the partial or complete mental image of a cat, or maybe the sound of a cat. Something cattish.

Now ask yourself: What color is the cat?

What next comes to mind?

For me, it was a gray cat and very soon afterward, the smell of kitty litter, which I intensely dislike. Then a memory of the laundry room in the house where I grew up, which is where my sister’s cat Samantha often slept.

The point is that anything that we focus on—an image seen or imagined, a song, a memory, and so on—causes the sensory areas of our brain to become active. That means our visual areas, auditory areas, the areas involved in processing touch, et cetera, all ramp up their levels of electrical activity. And while each of these areas has a different name and a different job in representing the world around us, they are all

extensively interconnected too. They don’t work in isolation. They work as parts of what are called distributed networks. Which sensory area is activated first and which ones follow depends on what “seeds” the thought. In the above example it was the word cat but think about the extremely soft fur of a chinchilla. That may set off a thought that activates the touch area of your brain, and you might experience a simulation of touch, what neuroscientists call somatosensation. From there, the patterns of brain activity expand.

Regardless of where it starts, focusing on something—anything—causes your brain to call up memories, as well as any potential motor actions you might take in relation to it. In some, it also conjures up emotions related to that thing. This “simulation” aspect of human brain function is vitally important. Not just as a way to navigate our lives, but for how we eventually come to make sense of our different life experiences. It is among our most valuable mental resources and a key feature of what sets us apart from the other species on earth.

As Groh explained, “Humans don’t hear or see any better than a macaque monkey. What we do have is more brain areas devoted to each of the same senses, which makes our brain capable of building up very elaborate simulations of what we perceive.” In other words, the more we focus on something the richer the simulation of that thing becomes.

That valuable mental resource is adaptive. It's what allows us to make predictions of "If A then B, and if B then maybe C" and so on. It's what allows us to imagine solutions to certain problems, and to build tools and technologies to overcome them. It's what gave us electricity, planes, the internet, smartphones, and so much more.

Our brains can only run a certain number of simulations at once. Groh described an experiment she does that illustrates this, and why. She stops one of her class lectures midway through and asks her students to try to think of a word totally unrelated to what they have been discussing. Sounds easy, right? Wrong. Eventually the students come up with a word, but it takes significantly more time and effort than if she asked them to think of a word related to what they have been discussing. It's as if thoughts are roads: The more time you spend thinking about something, the farther you travel down that road. Sounds obvious, but is it true? Yes. Neuroimaging and direct recordings of brain electrical activity confirm that as thought-triggered simulations get more elaborate, they outcompete other simulations.[11] Neuroscientists call these attractor states, which is when brain activity increases in a

[11] Jennifer M. Groh et al., "Signal Switching May Enhance Processing Power of the Brain," Trends in Cognitive Sciences 28, no. 7 (2024): 600–613, https://doi.org/10.1016/j.tics.2024.04.008.

certain set of circuits and as it does, activity in other circuits diminishes.[12]

Incidentally, this phenomenon may be what led to the idea that humans only use a small percentage of our brain at any one time—which unfortunately made it seem like we were dumber than we could be if only we used more of our brains. Now that you understand a bit about how thoughts work, it should be obvious why that's a silly idea. Who would want too many thoughts happening at once?!

Thought competition and attractor states are vital, in fact. Groh used a great example to explain why. Think about when you're driving and you say to someone, "Hold that thought for a second, I need to change lanes." You are reflexively limiting the amount of information coming in so you can switch simulations—from talking and listening, to

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[12] Guozhang Chen and Pulin Gong, "A Spatiotemporal Mechanism of Visual Attention: Superdiffusive Motion and Theta Oscillations of Neural Population Activity Patterns," Science Advances 8, no. 16 (2022): eabl4995, https://doi.org/10.1126/sciadv.abl4995; Edmund T. Rolls, "Attractor Network Dynamics, Transmitters, and Memory and Cognitive Changes in Aging," in Cognitive Changes and the Aging Brain, ed. Kenneth M. Heilman and Stephen E. Nadeau (Cambridge University Press, 2019), 203–25; Robert Englert et al., "Functional Connectivity-Based Attractor Dynamics of the Human Brain in Rest, Task, and Disease," eLife 13 (2026): RP98725, https://doi.org/10.7554/eLife.98725.3; Edmund T. Rolls, "Attractor Cortical Neurodynamics, Schizophrenia, and Depression," Translational Psychiatry 11, article no. 215 (2021), https://doi.org/10.1038/s41398-021-01333-7; Samming Song et al., "A Modular Latching Chain," Cognitive Neurodynamics 8 (2014): 37–46, https://doi.org/10.1007/s11571-013-9261-1.

driving and navigating. You are allocating resources to different brain circuits.

So maybe you're now asking, "What in the world does that mean for improving our relationship to ourselves?" Turns out: everything. Improving our relationship to ourselves is about being intentional with what thoughts we sow.

Gone unchecked, our thoughts grow, and grow, and grow. This can be a good or a bad thing for our mental and physical health and how we show up in the world.

It's important you know that your thoughts occur downstream of your perceptions. It's hard to overstate how strong this link really is. The way we see our lives, the way we narrate our lives and the events of the world at large to ourselves, shapes the thoughts we have. If you tell yourself that every time you try to do something new, you fail—well, your thoughts will reflect that perception. This realization is the first step in having control over them. So yes, I'm saying you can control your thoughts. Below I explain how. I promise none of what follows is about the power of positive thinking. I will not be guiding you through a series of manifestations. The first protocol in this section is, of course, a science-supported tool.

The other protocols that follow are about resolving a tension that exists for many people, regardless of age or stage of life, which is the push-pull between "needing to perform outward

roles in life" and "being true to oneself." That is one of the most common challenges I hear from people and it can produce varying degrees of confusion, conflict, and in some cases, crippling paralysis. I've often been asked by people forty and younger who are not already wholeheartedly committed to a career path, "What should I study, do, or be?" By contrast, people older than forty usually ask me about making meaning of their life.

With all the books and news articles and podcasts out there about finding a sense of purpose, I started to wonder: How can science help us answer these questions? It turns out there is a way. In my opinion, it's been best addressed by James Hollis, an eighty-six-year-old Jungian psychologist with more than five decades of clinical experience treating patients of all ages and backgrounds. He has helped them build their best life by taking specific steps daily. He's not a life coach, mind you. He's a credentialed therapist who has also authored twenty-two books on this subject. He was a guest on the Huberman Lab podcast in 2024, and I personally have used the tools he described then every day since our conversation to great benefit.[13] They resolve the tension between "needing to perform outward roles in life" and "being true to oneself." Hollis's "Daily Protocols for

[13] Andrew Huberman, "Dr. James Hollis: How to Find Your True Purpose & Create Your Best Life," Huberman Lab (podcast audio), May 13, 2024, https://www.hubermanlab.com/episode/dr-james-hollis-how-to-find-your-true-purpose-create-your-best-life.

Creating Your Best Life," as I refer to them in this chapter, are something I wish I had heard long before that conversation. They help us make decisions aligned with our interests, and reduce a lot of second-guessing that can arise when we consider such actions. No matter your age or level of contentment with your life, I am certain these protocols will be of great benefit for you too.

Finally, I include a protocol related to resolving inner conflicts that stem from unresolved pain. Formal trauma therapies are beyond the scope of this book and usually require working with a licensed practitioner, whether that's done in talk therapy, eye movement desensitization and reprocessing (often just called EMDR), hypnosis, psychedelics, or other modality. But there is a heavily researched approach to self-healing that has been shown time and time again to work in these cases, to lighten or even remove the chainmail these experiences cast upon us. It takes very little time, it's confidential, and requires no signing up for anything. It also costs nothing. It only requires that you are willing to confront the thing that is weighing on you. I provide the specific and most effective protocol for that here.

In short, what follows will set you up for more success in all the protocols listed up until now—and for whatever else you

decide to focus on and achieve in your life. It will improve your self-understanding in ways that unburden your mind.



PERSONAL GROWTH PROTOCOL

Replace Mind Wandering with Present-Tethered Awareness

WHAT TO DO

- Throughout your day, periodically ask yourself if your thoughts are focused on the present. For example: “Are my thoughts focused on what I am doing and where I am, or are they related to the past or future?” Be honest in your response.
- If your thoughts are elsewhere, re-tether them to the present. Ask questions anchored in your current surroundings, such as these: “How cold or warm is it in this room? What does it smell like? How many people in line in front of me are on their phone?” That seemingly trivial line of inquiry returns your perceptions to the present because they can only be answered right here, and now.

HOW IT WORKS

Most people would like to be happier. Among the many benefits, a better mood allows us to access more focus and creativity, enhances our rational assessment of information,

and improves countless objective health metrics.[14] On the flip side, psychologist Jamil Zaki's lab at Stanford has shown that a cynical mindset significantly limits people's problem-solving abilities, creativity, and physical health.[15] Similar reductions in problem-solving abilities and creativity have been shown among people with low mood.[16]

Fortunately, there is now good, actionable science on how to be happier and how to avoid unhappiness. Both involve—you guessed it—controlling your thoughts. But not by suppressing bad thoughts, or replacing them with positive ones. Quite the opposite in fact. Let me explain.

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[14] Radwa Khalil et al., "The Link Between Creativity, Cognition, and Creative Drives and Underlying Neural Mechanisms," Frontiers in Neural Circuits 13 (March 21, 2019): 18, https://doi.org/10.3389/fncir.2019.00018; Gerald L. Clore and Jeffrey R. Huntsinger, "How Emotions Inform Judgment and Regulate Thought," Trends in Cognitive Sciences 11, no. 9 (2007): 393–99, https://doi.org/10.1016/j.tics.2007.08.005; Kostadin Kushlev et al., "Does Happiness Improve Health? Evidence from a Randomized Controlled Trial," Psychological Science 31, no. 7 (2020): 807–21, https://doi.org/10.1177/0956797620919673.

[15] Jamil Zaki, Hope for Cynics: The Surprising Science of Human Goodness (New York: Grand Central Publishing, 2024).

[16] Bernard A. Nijstad et al., "The Dual Pathway to Creativity Model: Creative Ideation as a Function of Flexibility and Persistence," European Review of Social Psychology 21, no. 1 (2010): 34–77, https://doi.org/10.1080/10463281003765323; Arthur M. Nezu et al., "Social Problem Solving as a Risk Factor for Depression," in Risk Factors in Depression, ed. Keith S. Dobson and David J. A. Dozois (Amsterdam: Elsevier, 2008), 263–86; Nathan Ridout et al., "The Influence of Induced Dysphoria on Autobiographical Memory Specificity and Social Problem Solving: Examining the Role of Executive Function," Behaviour Research and Therapy 169 (October 2023): 104404, https://doi.org/10.1016/j.brat.2023.104404; C. Donaldson and D. Lam, "Rumination, Mood and Social Problem-Solving in Major Depression," Psychological Medicine 34, no. 7 (October 2004): 1309–18.

The 2010s marked an interesting shift in the Western perspective on happiness, and life goals generally. The change was catalyzed in part by a prominent study by Nobel Prize–winning psychologist Daniel Kahneman showing that beyond a certain threshold income level, having more money does not increase happiness.[17] (If you want to know, in 2010 that threshold amount was $75,000 annually—equal to about $110,000 today.) I confess, those results always gave me pause because let's face it: Money may not buy happiness, but it sure can help buffer against many common forms of stress, some obvious ones being housing, food, and healthcare. Turns out, there is a rarely mentioned aspect of the Kahneman study that found that more income did not increase daily happiness but it did improve people's overall sense of life satisfaction. Nevertheless, the notion that money can't buy happiness stuck around. I would argue one of the reasons for that is because it came at the same time as a rising interest in "mindfulness." That interest was accompanied by the idea that being "present" to our daily experiences, a mindset long embraced in Eastern traditions,

[17] Daniel Kahneman and Angus Deaton, "High Income Improves Evaluation of Life but Not Emotional Well-Being," Proceedings of the National Academy of Sciences 107, no. 38 (2010): 16489–93, https://doi.org/10.1073/pnas.1011492107.

is vital to our well-being. Just as important as material wealth, if not more so.[18]

The 2010s also brought a growing interest in the neural and psychological changes associated with mindfulness and also meditation, both of which are intended to help with being present. These practices are by no means new; they had existed in Eastern cultures for thousands of years, experienced fleeting popularity in the US in the 1970s, and then became trendy again. Now some fifteen to twenty years later, mindfulness is a familiar term, and meditation is a common practice. We also have a lot of data from lab studies and clinical trials showing that our level of "presence" is not just the outcome but a cause of happiness, both in the short and long term.[19] Okay, so we should "be present." Got it. But the whole "be present" thing is another pretty squishy term. Even for scientists wishing to study our level of presence, defining it is no easy task.

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[18] Anuradha Baminiwatta and Indrajith Solangaarachchi, "Trends and Developments in Mindfulness Research over 55 Years: A Bibliometric Analysis of Publications Indexed in Web of Science," Mindfulness 12, no. 9 (2021): 2099–116, https://doi.org/10.1007/s12671-021-01681-x.

[19] Laura G. Kiken et al., "Being Present and Enjoying It: Dispositional Mindfulness and Savoring the Moment Are Distinct, Interactive Predictors of Positive Emotions and Psychological Health," Mindfulness 8, no. 5 (2017): 1280–90, https://doi.org/10.1007/s12671-017-0704-3; E. S. Blanke et al., "Pathways to Happiness Are Multidirectional: Associations Between State Mindfulness and Everyday Affective Experience," Emotion 18, no. 2 (2018): 202–11, https://doi.org/10.1037/emo0000323.

To shift away from catchy phrases like “Be Here Now” (I love that phrase by the way) and run actual experiments, capture data, and suggest actionable tools on the basis of those data, scientists needed to create an operational definition of presence. They also needed to study it in the real world, which is not easy. Such studies were done by a few pioneering scientists to whom we owe a debt of gratitude, such as Richard Davidson, PhD, from the University of Wisconsin–Madison, and Matthew Killingsworth, PhD, and Dan Gilbert, PhD, from Harvard.

In 2010, Killingsworth and Gilbert published what is now considered a landmark study in Science.[20] First, they created an app to collect real-time information on participants’ thoughts and feelings as they went about their day. The app sent question prompts to people at random moments during waking hours, such as, “How are you feeling right now?” and “What are you doing right now?” Every prompt also included the question “Are you thinking about something other than what you’re currently doing?” The researchers amassed a database of nearly 250,000 responses from around 5,000 people ages eighteen to eighty-eight from eighty-three different countries.

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[20] Matthew A. Killingsworth and Daniel T. Gilbert, “A Wandering Mind Is an Unhappy Mind,” Science 330, no. 6006 (2010): 932, https://doi.org/10.1126/science.1192439.

Then they dove in. Killingsworth and Gilbert analyzed the responses from 2,250 adults, most from the US, to look for any connections between the specific activity they were doing, their level of presence with that activity, and their mood. Their results showed that for nearly 50 percent of our waking lives, our minds are elsewhere, thinking about times and places and things that have nothing to do with the present. But the researchers also showed the firm connection between presence and happiness. When study participants were focused on specific activities and on their immediate physical surroundings, they were happier. Even when those specific activities were boring or unpleasant, that focus was still a buffer against unhappiness. “A wandering mind is an unhappy mind,” the researchers wrote in a summary of their study.[21]

But perhaps the most interesting finding from the study was the causative link between mind wandering and unhappiness. When it came to feeling happy, mind wandering was a much bigger deterrent than whatever they were doing. It didn’t even matter what they were thinking about. Being present during a less enjoyable activity seemed to make people happier than daydreaming about pleasant things unrelated to the present activity. When we are focused on the present moment, we are happier. This is the power of present-tethered awareness.

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[21] Killingsworth and Gilbert, “Wandering Mind Is an Unhappy Mind.”

While that study has some obvious limitations—for a start, it's based on subjective self-reports—it is notable for a few reasons. First, it has a real-world context. It drew from large sample sizes. And it has readily implementable findings. It tells us that one of the most powerful determinants of our current and future levels of "happiness" is giving what we are doing as much of our undivided attention as we can. Notice I didn't say all our attention. Fully focusing 100 percent of our mental attention can happen, and that can be great—we discussed ways to do just that in the previous chapter. But it appears that the mere act of periodically devoting a bit more of your attention to the present can be of meaningful benefit to your level of happiness. The data tell us that striving to anchor your thoughts to the present forms the scaffolding of your well-being. So if you'd like to be happy, and you want your children to be happy, find ways to practice present-tethered awareness alone or together. The questions included in the protocol above are a good way to start.

Of course, feeling sad, disappointed, angry, et cetera, is sometimes warranted. It's also healthy.[22] Suppressing negative feelings has consistently been shown to lead to

[22] B. Q. Ford et al., "The Psychological Health Benefits of Accepting Negative Emotions and Thoughts: Laboratory, Diary, and Longitudinal Evidence," Journal of Personality and Social Psychology 115, no. 6 (2018): 1075–92, https://doi.org/10.1037/pspp0000157.

poorer mental and physical health outcomes.[23] The problem is when we get stuck in those low-mood states too long.

Studies that followed supported general findings from this landmark study but brought some important revisions too. There are benefits to letting your mind wander too. Daydreaming during familiar activities that don't require much attention generally supports our well-being and creativity. As someone who daydreams during cleaning tasks, long drives, and hikes, I find that reassuring. It also helps explain why people report getting some of their best insights while in the shower, while doing routine tasks such as washing dishes or mowing the lawn, while on walks or commuting, and so on. A 2017 analysis by Shaghayegh Konjedi and Reza Maleeh confirmed, however, that mind wandering during emotionally or physically challenging negative experiences was detrimental to people's immediate and future mood, just as the earlier study had concluded.[24]

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[23] Benjamin P. Chapman et al., "Emotion Suppression and Mortality Risk over a 12-Year Follow-Up," Journal of Psychosomatic Research 75, no. 4 (2013): 381–85, https://doi.org/10.1016/j.jpsychores.2013.07.014; Alexandra T. Tyra et al., "Emotion Suppression and Acute Physiological Responses to Stress in Healthy Populations: A Quantitative Review of Experimental and Correlational Investigations," Health Psychology Review 18, no. 2 (2024): 396–420, https://doi.org/10.1080/17437199.2023.2251559; Jainish Patel and Prittesh Patel, "Consequences of Repression of Emotion: Physical Health, Mental Health and General Well-Being," International Journal of Psychotherapy Practice and Research 1, no. 3 (2019): 16–21, https://doi.org/10.14302/issn.2574-612X.ijpr-18-2564.

[24] Shaghayegh Konjedi and Reza Maleeh, "A Closer Look at the Relationship Between the Default Network, Mind Wandering, Negative Mood, and

The takeaway is clear: Whenever you can, bring your thoughts back to what you are doing. Don't chastise yourself for letting your mind wander but understand its negative impact on your mood. Use the simple recommendation above to re-tether your mind to the present.

If you tend to daydream, try to do so during activities that don't require much focus. In today's world, where the first seconds of waiting in line at the store, airport, or a doctor's office has us immediately reaching for our smartphone to text or scroll, a bit of good old-fashioned daydreaming is probably a better idea.

Getting Off Your Phone

Is browsing social media and the internet violating present-tethered awareness?

As that 2010 study discussed above indicates, people's minds were already wandering about 50 percent of their waking hours before social media became such a prominent—arguably even dominant—feature of modern life that it is now. Tech companies just figured out how to get people to devote a significant percentage of that time to scrolling on their social media platforms.

Depression," Cognitive, Affective, & Behavioral Neuroscience 17, no. 4 (2017): 697–711, https://doi.org/10.3758/s13415-017-0506-z.

My view is that, in moderation, meaning about an hour or so per day, engaging on social media platforms can be great for learning, social interaction, and entertainment. The key is to not use social media to blunt your present-tethered awareness. We are starting to see the damage to mental and physical health that this can cause.[25] Put simply, scrolling can become just another form of mood-lowering mind wandering. The algorithms are strong. Their superpower is to get you to use them for as long as possible. Your superpower is your ability to consciously redirect your thoughts to specific activities and your immediate physical surroundings. That's a skill worth developing. If the benefits of doing so for your mood and well-being are not incentive enough, perhaps the longevity benefits I'll describe next will be.

One of the causes of aging is the shortening of our telomeres, small molecular structures that form protective caps on the ends of our chromosomes in each of our cells. In cell types that duplicate, such as our immune cells, as telomeres get shorter, they stop dividing. At that point they become what are called zombie cells, and they secrete inflammatory

[25] Osman Ulvi et al., "Social Media Use and Mental Health: A Global Analysis," Epidemiologia 3, no. 1 (2022): 11–25, https://doi.org/10.3390/epidemiologia3010002; Bdour Alwuqaysi, "A Systematic Review and Meta-Analysis of Social Media's Influence on Mental Health, Family Functioning, and Visual Content," Social Sciences & Humanities Open 12 (2025): 101793, https://doi.org/10.1016/j.ssaho.2025.101793.

molecules that damage nearby cells. Preclinical studies suggest that this zombie cell–induced damage may be a factor in aging and age-related illnesses such as cancer, dementia, and heart disease.[26] Conversely, having cells with longer telomeres is a biomarker of youthful, healthy, and (for cell types that replicate) division-competent cells.[27]

With that knowledge in hand, UCSF psychologist Elissa Epel and colleagues conducted a study where they measured the average telomere length of immune cells from women who scored either high, moderate, or low on measures of mind wandering.[28] She discovered that the women engaged in the greatest amount of task-irrelevant thought throughout the day had telomeres that were on average two hundred base pairs (a base pair is a functional unit of DNA) shorter than the telomeres of women with low amounts of mind wandering. We all lose about 50 telomere base pairs for each year of life, so the difference between being someone who is present and someone whose mind wanders was equivalent to

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[26] Diana Kwon, "How Your Brain Controls Ageing—and Why Zombie Cells Could Be Key," Nature 642, no. 8068 (2025): 563–65, https://doi.org/10.1038/d41586-025-01886-3; Ming Xu et al., "Senolytics Improve Physical Function and Increase Lifespan in Old Age," Nature Medicine 24, no. 8 (2018): 1246–56, https://doi.org/10.1038/s41591-018-0092-9.

[27] Enzo Tedone et al., "Telomere Length and Telomerase Activity in T Cells Are Biomarkers of High-Performing Centenarians," Aging Cell 18, no. 1 (2019): e12859, https://doi.org/10.1111/acel.12859.

[28] Elissa S. Epel et al., "Accelerated Telomere Shortening in Response to Life Stress," Proceedings of the National Academy of Sciences 101, no. 49 (2004): 17312–15, https://doi.org/10.1073/pnas.0407162101.

about four years of additional aging. This was an especially remarkable finding because the benefits appeared independent of variables known to impact longevity, such as body mass index and stress level. The takeaway: Make the effort to be present in your daily activities, tasks, and relationships. Doing so provides an actionable way to improve your mental and physical health and, potentially, your lifespan.

This doesn't mean you have to be a Luddite. Texting a friend, sending a work email, or updating a family schedule on your phone can definitely be about the "here and now." So is scrolling for education or entertainment. The thing to be wary of is texting and scrolling or doing anything out of boredom or avoidance. Ask yourself: "Am I numbing out or getting drawn in by things that I can't actually have an impact on?" If you answer yes, that's a good indicator that you are engaged in deleterious mind wandering. Retired Navy SEAL and two-time world-record-holding athlete Andy Stumpf talks about this in his book Drownproof.[29] And he shared a tool that I have found immensely useful, even life-changing. Here it is: Take a sheet of paper and draw a line down the middle from top to bottom. On the left side, list all the things that concern you at that moment. On the right side, list all the things you can actually control, and

[29] Andy Stumpf, Drownproof: Eight Life Lessons to Keep Your Head Above Water (St. Martin's Press, 2026).

what you can do to directly impact your life that day, right then and there. Try it. You'll be amazed at what you learn about your mental preoccupations and reflexes, and what it opens up in terms of your ability to better direct your mind and actions. Doing this exercise makes you keenly aware of what really matters to you. And it gives you a heightened sense of agency around those things. It places you in a great mental stance to lean into the important stuff in your life. It also makes time spent relaxing on the couch or doing nothing feel restorative again. It's a simple tool, but a super valuable one. Yes, it's a good idea to do it every few weeks or once a month, but it's quick: five minutes at most. Do it.

Getting Out of Your Head

I am a fan of certain forms of structured introspection. I'm referring to journaling, therapy with a well-trained expert who offers useful points of action, or other deliberate, timed self-reflection done within a specific framework. There are very good data to support those practices as tools for mental health.[30] What none of us want, however, is to be stuck in our own heads and feeling anxious. Turns out, we don't have to. We just need to understand a bit more neuroscience. And then apply that knowledge.

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[30] Monika Sohal et al., "Efficacy of Journaling in the Management of Mental Illness: A Systematic Review and Meta-Analysis," Family Medicine and Community Health 10, no. 1 (2022): e001154, https://doi.org/10.1136/fmch-2021-001154; Wang He and Jun Gan, "The

Your brain has cells that are grouped into circuits, which in turn are grouped into networks. Think of a network sort of like a setting for your brain: You have sleep networks, feeding networks, language networks, math networks, and on and on. And you have a network that turns on when you do nothing, aptly named the default mode network, or DMN. It was discovered to be active when people were getting their brains scanned, in between moments where they were asked to focus on certain tasks.[31] That alone wasn't terribly surprising. What was surprising was that the DMN is hyperactive in ruminators, people with cognitive impairments like Alzheimer's disease, and in depressed patients.[32] More than a decade of studies now show that many antidepressants work in part by dampening activity in

Relationship Between Self-Reflection and Mental Health: A Meta-Analysis Review," Current Psychology 44 (2025): 3899–913, https://doi.org/10.1007/s12144-025-07415-9; Christoph Flückiger et al., "Psychotherapy Works—an Inclusive and Affirming View to a Modern Mental Health Treatment," Clinical Psychology in Europe 6 (April 2024): e11971, https://doi.org/10.32872/cpe.11971.

[31] Vinod Menon, "20 Years of the Default Mode Network: A Review and Synthesis," Neuron 111, no. 16 (2023): 2469–87, https://doi.org/10.1016/j.neuron.2023.04.023; Wei Luo et al., "Rest to Promote Learning: A Brain Default Mode Network Perspective," Behavioral Sciences 14, no. 4 (2024): 349, https://doi.org/10.3390/bs14040349.

[32] Hui-Xia Zhou et al., "Rumination and the Default Mode Network: Meta-Analysis of Brain Imaging Studies and Implications for Depression," NeuroImage 206 (February 1, 2020): 116287, https://doi.org/10.1016/j.neuroimage.2019.116287; Sahar Riaz et al., "Functional Connectivity in the Default Mode Network During Rumination in Depression: A Systematic Review," BJPsych Open 11, no. S1 (2025): S62–63, https://doi.org/10.1192/bjo.2025.10214; L. L. Beason-Held, "Dementia and the Default Mode," Current Alzheimer Research 8, no. 4 (2011): 361–65,

the DMN.[33] Guess what? So does dancing, exercising, socializing with friends, drawing… essentially anything that requires your attention.[34] That's the key takeaway about the DMN. You can't just "turn it off" or even reduce its activity unless you do something to activate a competing brain network, such as the salience network—a different collection of circuits that, you guessed it, become active when you focus your attention on something external to you.[35]

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https://doi.org/10.2174/156720511795745294; Vincent Malotaux et al., "Default-Mode Network Connectivity Changes During the Progression Toward Alzheimer's Dementia: A Longitudinal Functional Magnetic Resonance Imaging Study," Brain Connectivity 13, no. 5 (2023): 287–96, https://doi.org/10.1089/brain.2022.0008; Tina Chou et al., "The Default Mode Network and Rumination in Individuals at Risk for Depression," Social Cognitive and Affective Neuroscience 18, no. 1 (2023): nsad032, https://doi.org/10.1093/scan/nsad032.

[33] Jonathan Posner et al., "Antidepressants Normalize the Default Mode Network in Patients with Dysthymia," JAMA Psychiatry 70, no. 4 (2013): 373–82, https://doi.org/10.1001/jamapsychiatry.2013.455; Guido A. van Wingen et al., "Short-Term Antidepressant Administration Reduces Default Mode and Task-Positive Network Connectivity in Healthy Individuals During Rest," NeuroImage 88 (March 2014): 47–53, https://doi.org/10.1016/j.neuroimage.2013.11.022.

[34] Marjo Kullberg-Turtiainen et al., "Individualized Goal Directed Dance Rehabilitation in Chronic State of Severe Traumatic Brain Injury: A Case Study," Heliyon 5, no. 2 (2019): e01184, https://doi.org/10.1016/j.heliyon.2019.e01184; Young-woo Ko et al., "Changes in Functional Connectivity Between Default Mode Network and Attention Network in Response to Changes in Aerobic Exercise Intensity," Psychiatry Investigation 20, no. 1 (2023): 27–34, https://doi.org/10.30773/pi.2022.0245; Yaara Yeshurun et al., "The Default Mode Network: Where the Idiosyncratic Self Meets the Shared Social World," Nature Reviews Neuroscience 22, no. 3 (2021): 181–92, https://doi.org/10.1038/s41583-020-00420-w.

[35] Cátia Ribeiro da Costa et al., "Interplay Between the Salience and the Default Mode Network in a Social-Cognitive Task Toward a Close Other," Frontiers in Psychiatry 12 (February 6, 2022): 718400, https://doi.org/10.3389/fpsyt.2021.718400.

So while your DMN has important healthy functions, like simulating what other people are thinking or feeling, which is useful for some kinds of social interactions, to stop ruminating and to alleviate depressive states, you want to actively inhibit your DMN. Practicing present-tethered awareness is the way to do that too. If you are too much in your own head, feeling anxious or depressed, you would do well to focus outside yourself, literally on your surroundings. Better yet, engage in an activity that forces you to do that, such as exercising, reading a book, or listening to a podcast or music. If you feel your mind is still looping even while doing those things and that you can't seem to turn off that dominant inner narrative, try the exercise Andy Stumpf wrote about. Of course, clinical major depression deserves professional support. But for anyone seeking to be happier, more fulfilled, and more on track in life, learning to recognize what thoughts are, understanding how we can direct them, and then directing them to the present, especially toward the things in the present we can have a positive impact on… that's the ticket to a great life.



PERSONAL GROWTH PROTOCOL

Apply James Hollis's Guide to Creating Your Best Life

WHAT TO DO: Practice these two things every single day.

- 1 Apply the Shut Up. Suit Up. Show Up motto to your everyday life. These are science-based tools to help bring your outward and inner lives into alignment. Shut Up: Don't whine, instead incorporate a daily gratitude practice into your routine.
- Suit Up: Use a task-switching protocol to move between tasks associated with your outward roles.
- Show Up: Fulfilling our outward roles moves our life forward. That is what gives you real confidence.
- 2 Allow your internal maps to update. The best way to do this is to exit what neuroscientists and psychologists call stimulus-response mode. Here are two ways to do so: Engage in an activity unrelated to any ambitions or outward roles. Draw, read, think, stare at the sky, go biking… whatever it is, this is just for you.
- Simply close your eyes and let your thoughts emerge. Practice not doing anything about them. At least not in those moments. While this sounds passive, it allows your mind to reconnect to what's most important to you.

HOW IT WORKS

A great outward life requires that we show up and perform well in roles at school and work, and in our relationships. That requires we handle our responsibilities, even when we don't feel like it. A great inner life requires that we pursue the roles at school and work and in our relationships that are right for us, not just out of duty. When our outward and inner lives are aligned, life is good. When they conflict, things take more energy. We second-guess ourselves. We ruminate. We feel more tired from less effort. Psychologist James Hollis has developed a simple way for us to understand, and integrate, our outward and inner lives. It's a daily practice. It's for everyone, regardless of individual differences. It's applicable in both good and hard times. It's good, actionable science on how to be happier and how to avoid wasting valuable years of your life.

The key, as Hollis says, is to never expect your best self to emerge by default. Hollis has a simple motto he recites to himself every morning, and throughout the day. He recommends his patients write it down and carry it with them, and recite it to themselves every morning, and throughout the day too.

This protocol is named after his motto: "Shut Up. Suit Up. Show Up." While that might sound harsh, it's not what it seems. The motto is the inspiration for a collection of

science-based tools that can help you build your best life. Let me explain.

Shut Up

This means don’t whine; there is always someone suffering more than you. Instead, adopt a daily gratitude practice—and do it the way the science has shown to be effective (more on that below). It’s especially important to avoid whining and practice gratitude when you are feeling overwhelmed or down. I know many people are probably thinking, “So I’m supposed to suppress my feelings?” Hollis is a psychologist not a coach. He’s all about acknowledging feelings. But he also is aware of what it takes to build a life that brings a person lasting joy. So stay with me.

Suit Up

This means be prepared. Take the steps you can each day to arrive at your outward roles as best you are able. This sometimes looks like self-care: Get adequate sleep and don’t overbook yourself. It can also look like work: Finish your assignments, be on time, keep your belongings in order, schedule your kids’ activities, and so on. Homework doesn’t end when we finish school.

Show Up

This means hold yourself accountable for your work, for how you act in relationships, and indeed for all the outward roles you take on in life. You are in the driver's seat, so to speak.

Now before you tell me all that just sounds like Boomer diligence, consider this: Our inner narratives about our self-worth, and our self-esteem, are strongly impacted from fulfilling these outward roles.[36] Now is that the whole picture? Of course not. There are many factors at play when it comes to how we talk and think and feel about ourselves. But this is a significant fraction of it. Alas, just telling yourself, "You are enough just as you are," won't lead you to believe it… unless you are fulfilling these outward roles. This is what Hollis points out. If you're a member of a sports team, you show up to practice and work hard, with the subsequent feeling of pride in yourself. If you're a parent, you work hard at doing right by your child, with the subsequent satisfaction of having given that your all.

Holding ourselves accountable to our chosen roles also leads us into a state of thankfulness for the opportunity to perform well in them. Hollis explained to me that this state is the most important aspect of the practice. We choose a role, we

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[36] Trenton D. Mize and Reilly Kincaid, "Role-Accumulation and Mental Health Across the Life Course," American Sociological Review 90, no. 2 (2025): 226–56, https://doi.org/10.1177/00031224241313394.

commit to it, we do well, and then we are lifted into gratitude. Giving our roles our best each day becomes an act of gratitude—the act of living our life becomes a devotional practice of being thankful. Sound corny? Perfect! Because it's not about saying it. It's about the actions associated with each of the sentences. Let's get to it.

Shut Up, Redefined

At first glance, you might think that "shut up" means stop complaining. But what Hollis is after here is replacing complaining with something more beneficial. And the suggestion is more nuanced than simply "feel grateful." Rather, try spending a short amount of time each day thinking about something that causes you to feel grateful that you would not be aware of without some effort. This is not a simple gratitude list. It's about searching for something we would otherwise be unaware of or take for granted. Take a minute, or five. The data tell us this dedicated effort is positively transformative for your health, and more.[37]

In many ways, gratitude is a lot like hydration, but for your emotional well-being. Its importance for basic daily functioning is overlooked. People think of it as an extra, but

[37] Robert A. Emmons and Michael E. McCullough, "Counting Blessings Versus Burdens: An Experimental Investigation of Gratitude and Subjective Well-Being in Daily Life," Journal of Personality and Social Psychology 84, no. 2 (2003): 377–89, https://greatergood.berkeley.edu/pdfs/GratitudePDFs/6Emmons-BlessingsBurdens.pdf.

it's foundational to high performance. Of all the things I've covered on my podcast, this is the tool that surprised me the most in terms of its potency. Thinking back to what we now know about thoughts—what they are and how they grow and compete with one another—it makes sense. When we are seeding thoughts of gratitude, we suppress negative thoughts. We also run simulations from a place of security and stability, rather than a place of lack. Much more feels possible from that place. So forget the idea that if you do a gratitude practice daily you'll become complacent. Actually, the opposite happens. Even a short gratitude practice of a few minutes has been shown to elevate motivation and mood, and can even improve memory recall and cognitive ability.[38]

Gratitude tends to increase other positive feelings and states, like happiness and optimism too.[39] And there is evidence that this link is causative. In a 2003 study by psychologists Robert Emmons and Michael McCullough, 192 undergraduates were randomly assigned to one of three

[38] Yukako Tani et al., "Association Between Gratitude, the Brain and Cognitive Function in Older Adults: Results from the NEIGE Study," Archives of Gerontology and Geriatrics 100 (May–June 2022): 104645, https://doi.org/10.1016/j.archger.2022.104645; Sunghyon Kyeong et al., "Effects of Gratitude Meditation on Neural Network Functional Connectivity and Brain-Heart Coupling," Scientific Reports 7, no. 1 (2017): 5058, https://doi.org/10.1038/s41598-017-05520-9.

[39] M. E. McCullough et al., "The Grateful Disposition: A Conceptual and Empirical Topography," Journal of Personality and Social Psychology 82, no. 1 (2002): 112–27, https://doi.org/10.1037/0022-3514.82.1.112.

groups, each of which was asked to reflect on the past week by writing about (1) up to five things they were grateful for; (2) up to five things that were a hassle (a dirty kitchen, doing poorly on a test, etc.); or (3) five events that affected them in some way in the past week (attending a concert, cleaning a closet, and so on).[40] All three groups also rated their mood, any physical symptoms they were experiencing, such as headaches, congestion, sore muscles, nausea, skin irritation, poor appetite, or other minor issues, and how they reacted to any offers of help over the past week. A second study followed the same design but with daily writing instead of weekly. And a third study randomly assigned people with neuromuscular disease to either the first writing group or a control group that did only the self-rating part of the study.

All three studies—weekly reflecting, daily reflecting, and reflecting among people coping with a serious illness—showed that deliberately exercising gratitude was associated with more optimism, fewer minor health issues, more exercising, a greater tendency to lend emotional support to others, and other benefits. Among the participants with neuromuscular disease, those in the gratitude writing group slept better and felt more connected to others. Importantly, the data analysis showed that these outcomes were the specific result of inducing gratitude in oneself and not a side effect of the study as a whole.

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[40] Emmons and McCullough, “Counting Blessings Versus Burdens.”

Gratitude is also good for the heart. One study of eighty-six adults looked at changes in blood pressure and heart rate in response to a stimulus and alongside participants' responses to questions intended to gauge their level of gratitude.[41] The researchers found no connection between trait gratitude and cardiovascular reactivity. However, participants with higher levels of state gratitude tended to not experience the same spikes in blood pressure that the other participants did. A review of nineteen studies, including a total of 2,951 participants, found links between gratitude and a lower risk of cardiovascular disease and a greater likelihood of healthy behaviors, such as maintaining a good diet and adhering to prescription medication.[42]

It's important to note this refers to what psychologists call state gratitude, as opposed to trait gratitude. The latter is a personality trait. Some people have a disposition that is just naturally more grateful.[43] And yes, they do better in life.[44]

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[41] Stephen Gallagher et al., "State, but Not Trait Gratitude Is Associated with Cardiovascular Responses to Acute Psychological Stress," Physiology & Behavior 221 (July 1, 2020): 112896, https://doi.org/10.1016/j.physbeh.2020.112896.

[42] Xiaoxiao Wang and Chunli Song, "The Impact of Gratitude Interventions on Patients with Cardiovascular Disease: A Systematic Review," Frontiers in Psychology 14 (September 20, 2023): 1243598, https://doi.org/10.3389/fpsyg.2023.1243598.

[43] McCullough et al., "Grateful Disposition."

[44] Gewnhi Park et al., "The Benefit of Gratitude: Trait Gratitude Is Associated with Effective Economic Decision-Making in the Ultimatum Game," Frontiers in Psychology 12 (April 19, 2021): 590132, https://doi.org/10.3389/fpsyg.2021.590132; Kai Zhao and Wenna Ji, "The

State gratitude, meanwhile, is about appreciation for what we are given. For most people, state gratitude needs to be cultivated intentionally. It does not emerge by default.

Some data even suggest a link between gratitude and mortality. A study published in JAMA Psychiatry in 2024 looked at data on 49,275 female registered nurses from across the US, obtained through the Nurses' Health Study, which has been ongoing since 1976.[45] Specifically, the researchers, from Harvard and the University of British Columbia, focused on responses to a six-item questionnaire about gratitude that enrollees first completed in 2016 and the 4,608 deaths that occurred between 2016 and 2019. The analysis showed a link between gratitude and mortality: The greater the gratitude, the less likely a person was to die during the time frame of the study. "This study provides the first empirical evidence suggesting that experiencing grateful affect is associated with increased longevity in older adults," the authors wrote.

The benefits of this practice are astounding, but if you're curious as to how to actually feel gratitude, psychologist Robert Emmons, professor emeritus at University of

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Influence of Gratitude and Personality Traits on Career Resilience and Career Success Among College Students," Frontiers in Psychology 15 (April 23, 2024): 1340200, https://doi.org/10.3389/fpsyg.2024.1340200.

45 Ying Chen et al., "Gratitude and Mortality Among Older US Female Nurses," JAMA Psychiatry 81, no. 10 (2024): 1030–38, https://doi.org/10.1001/jamapsychiatry.2024.1687.

California, Davis, and a pioneer in the research of gratitude and positive psychology, tells us how. He found that gratitude comes from a two-step cognitive process:

(a) recognizing that one has obtained a positive outcome [in some area of life], and

(b) recognizing that there is an external source for this positive outcome.

Cultivating gratitude requires commitment. So Hollis's recommendation of pausing each day to acknowledge what we are grateful for is a logical starting point. But weaving together some compelling research suggests ways to make this protocol even more effective.

First, pioneering neuroscientist Antonio Damasio and colleagues investigated brain regions associated with gratitude by asking participants to listen to stories from genocide survivors who received lifesaving assistance from strangers and to imagine how they would feel about that care, rating their level of gratitude for each incident of assistance.[46] The researchers found that participants documented feelings of gratitude correlated with brain activity in the anterior cingulate cortex and the medial prefrontal cortex, regions associated with values, morals, and

[46] Glenn R. Fox et al., "Neural Correlates of Gratitude," Frontiers in Psychology 6 (September 29, 2015): 1491, https://doi.org/10.3389/fpsyg.2015.01491.

theory of mind (which refers to the ability to understand what another person might be thinking). The approach taken in this study is important. Note that the participants here were not thinking about their own lives. They were putting themselves in other people's shoes.

The final clue here comes from a study in which participants wrote letters of gratitude to their coworkers, unbeknownst to that coworker.[47] When the coworkers received these letters, researchers were on call. They took brain images while the coworkers listened to what had been written about them. Like in the prior study, hearing these feelings of gratitude correlated with prefrontal cortex activation, suggesting that the recipients were experiencing their own gratitude.

All of these findings point toward the importance of practicing gratitude. Make it nonnegotiable for a few minutes each day. Try Hollis's method. Or put yourself in the mindset of others experiencing feelings of gratitude through story or solicit examples of when your friends, family, and colleagues felt grateful for you.

You can also seek out stories that are powerful for you—such as in documentaries or in books or movies—where you observe someone sharing gratitude for actions performed by others. This approach is likely to be as

[47] Daisuke Hori et al., "Prefrontal Activation While Listening to a Letter of Gratitude Read Aloud by a Coworker Face-to-Face: A NIRS Study," PLOS One 15, no. 9 (2020): e0238715, https://doi.org/10.1371/journal.pone.0238715.

effective as listing what you personally are grateful for, as is often recommended in journaling practices. It is aligned with how our brains link observing gratitude to our internal sense of reward.[48] We are clearly wired to feel better from observing and receiving genuine gratitude.

Suit Up, Redefined

"Suit up" may sound somewhat, well, obvious: We all know to shower, get dressed for school or work, pack our lunch and perhaps the kids' too, and get on with the day. But this isn't about any of that. It's about what neuroscientists call task switching.

Task switching is an essential aspect of role fulfillment but is rarely discussed. Moving our attention from one chore or job or activity to another is fundamental to being effective in whatever outward roles we perform. Good task switchers are more focused and they are calmer when things get busy.[49] They are often the kind of people about which others say, "I don't know how they do it all, much less so gracefully…"

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[48] Hori et al., "Prefrontal Activation."

[49] Tobias Egner, "Principles of Cognitive Control over Task Focus and Task Switching," Nature Reviews Psychology 2, no. 11 (2023): 702–14, https://doi.org/10.1038/s44159-023-00234-4; Lena Rademacher et al., "Individual Differences in Resilience to Stress Are Associated with Affective Flexibility," Psychological Research 87, no. 6 (2023): 1862–79, https://doi.org/10.1007/s00426-022-01779-4.

Neuroscience has a lot to say about task switching, and it has everything to do with our earlier discussion about thoughts. As we spend more time doing a task, and the internal simulations around that task get more and more elaborate, a greater and greater switch cost accumulates.[50] That means that it gets harder to move on from that task, and those internal simulations, to a new task, and new internal simulations. Switch costs are real and significant, reducing our mental and physical performance by anywhere from 15 to 40 percent. Sometimes it's a slowing of response times, other times it's a reduction in accuracy, or both.[51] Think of it this way: Doing something for a long time gets you deep into a neural trench, so to speak, and getting out of that trench requires more time and more energy. Some people seem to intuitively know about this aspect of brain function, and always add a few minutes between tasks to get out of a neural trench. Others of us make the false assumption that we can hop between tasks and be at our best. You know who doesn't? Athletes, cognitive athletes (like professional chess

[50] J. S. Rubinstein et al., "Executive Control of Cognitive Processes in Task Switching," Journal of Experimental Psychology: Human Perception and Performance 27, no. 4 (2001): 763–97, https://doi.org/10.1037/0096-1523.27.4.763.

[51] Michèle C. Muhmenthaler and Beat Meier, "Task Switching Hurts Memory Encoding," Experimental Psychology 66, no. 1 (2019): 58–67, https://doi.org/10.1027/1618-3169/a000431; Kamrul Hasan, "Digital Multitasking and Hyperactivity: Unveiling the Hidden Costs to Brain Health," Annals of Medicine & Surgery 86, no. 11 (2024): 6371–73, https://doi.org/10.1097/MS9.0000000000002576.

players), and creative artists. They are famed for their stereotyped rituals around both practice and competitions. They know how to "get into frame," "reset between points," and on and on.[52] Six-time Mr. Olympia winner Dorian Yates—known for his brief but extremely intense training (which, by the way, was based in large part on Mike Mentzer's principles I described in chapter 2)—told me he ironed his clothes prior to every workout… not just for the appearance but as a mental preparation ritual for what was to come. He was actively steering his mind.[53] Loads of books on high performance have delved into such practices and found that, when it comes to routines to harness our mental and physical resources so we can apply them with the utmost focus and effectiveness, what works best is what can be ritualized, not necessarily the specific thoughts or actions.[54] It's not that you take three slow breaths emphasizing the exhale to slow your heart rate or intentionally blink fast five

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[52] Simone Montuori et al., "Executive Functioning Profiles in Elite Volleyball Athletes: Preliminary Results by a Sport-Specific Task Switching Protocol," Human Movement Science 63 (February 2019): 73–81, https://doi.org/10.1016/j.humov.2018.11.011; Enrico Premi et al., "Enhanced Dynamic Functional Connectivity (Whole-Brain Chronnectome) in Chess Experts," Scientific Reports 10, no. 1 (2020): 7051, https://doi.org/10.1038/s41598-020-63984-8.

[53] Andrew D. Huberman, "Dorian Yates: Build Muscle & Strength & Forge Your Life Path," Huberman Lab (podcast audio), January 19, 2026, https://www.hubermanlab.com/episode/build-muscle-and-strength-and-forge-your-life-path-dorian-yates.

[54] As one example, see the following: Leon Bax, Building Performance Routines: Empowering Habits Toward Impact (2Mynds, 2025).

times before you sit down to work, it's that you do something that you can do most anywhere, anytime and that you have decided is your ritual. Pick one, and design your own task-switching ritual.

Studies across a range of sports and cognitive tasks, like public speaking or test taking, show that stereotyped physical actions are the most effective task-switching rituals, as opposed to just mental mantras or self-talk. Think of the way a baseball player taps the bat on home plate before the first pitch. These may "shake off" the neural residue of the thoughts and actions related to your prior activity by shutting down the specific circuits involved in it. This is a distributed phenomenon, which means it does not happen in a single region of the brain, but it does critically involve your basal ganglia, which control "go" circuits that generate action commands for your brain and body, as well as "no go" circuits responsible for suppressing specific actions.[55]

Your own short task-switching ritual will make you more effective at moving from one task to the next, which has outsized positive benefits in all your outward roles.

So when Hollis talks about "suiting up," and getting prepared for your outward roles, or when former Navy SEAL Jocko Willink talks about putting out your workout

[55] Michael Falkenstein et al., "Action Monitoring, Error Detection, and the Basal Ganglia: An ERP Study," NeuroReport 12, no. 1 (2001): 157–61, https://doi.org/10.1097/00001756-200101220-00039.

clothes the night before an early morning session,[56] or when the great choreographer Twyla Tharp talks about ritualizing the transition into and out of her legendarily long and repetitive dance practices, they are all talking about the same thing: the importance of having a ritualized task-switching protocol.[57]

To be an effective task switcher, you also need to not do certain things. Nowadays we fill every second between activities—sets in the gym, the wait for food to arrive at a restaurant, the drive to pick up the kids from school, and so on—with smartphone use. Then we wonder why we are struggling to manage it all. We are overscheduled, overstimulated, and overtired. On days when we do sleep well, our task-switching abilities are vastly improved, but most of us need, and can benefit from, "anytime" actions to help us task switch to be ready and at our best for when we "show up."

Show Up, Redefined

[56] Richard Feloni, "A Retired Navy SEAL Commander Does 2 Things Every Night Before Bed So He Can Attack the Next Day the Minute He Wakes Up," Business Insider, November 10, 2017, https://finance.yahoo.com/news/retired-navy-seal-commander-does-141500339.html.

[57] Twyla Tharp, The Creative Habit: Learn It and Use It for Life (Simon & Schuster, 2006).

This task-switching protocol can add benefit to our cognitive, emotional, and physical capabilities.[58] And yes, many of the things that you may already do—taking a few long exhale breaths as you return home at the end of the day, or before getting into work mode when you arrive at the office or into workout mode when you step inside the gym—are about this whole notion.

So make it work for you. And remember that the ritual between tasks associated with different outward roles can be exactly the same. Same for every switch. It's yours; do whatever works. This tool, as I said above, places the locus of control inside us. It moves us away from the need for conditions outside us to always be optimal to tackle our outward roles. It puts us in charge of how we start our tasks, and gives us confidence to continue.

I'm guessing that when you first read Shut Up. Suit Up. Show Up., you probably weren't thinking about gratitude practices, or about your basal ganglia, but what makes

[58] Tobias Egner and Audrey Siqi-Liu, "Insights into Control over Cognitive Flexibility from Studies of Task-Switching," Current Opinion in Behavioral Sciences 55 (February 2024): 101342, https://doi.org/10.1016/j.cobeha.2023.101342; Rademacher et al., "Individual Differences in Resilience to Stress"; Bettina Wollesen et al., "Effects of Cognitive-Motor Dual Task Training on Cognitive and Physical Performance in Healthy Children and Adolescents: A Scoping Review," Acta Psychologica 224 (April 2022): 103498, https://doi.org/10.1016/j.actpsy.2022.103498; Laurence Desjardins-Crepeau et al., "Effects of Combined Physical and Cognitive Training on Fitness and Neuropsychological Outcomes in Healthy Older Adults," Clinical Interventions in Aging 11 (September 2016): 1287–99, https://doi.org/10.2147/cia.s115711.

Hollis's approach uniquely powerful is that it balances introspection with action. It's neither coddling nor harsh. And it's realistic to the demands of life we all have to meet, as well as the internal challenges most all of us feel at some point or another. I take comfort in the fact that there are science-backed tools to help foster Shut Up. Suit Up. Show Up.

We're an oddly fortunate species. On the one hand we are born with unmatched cognitive capacity, as well as the ability, through neuroplasticity, to continually learn. No other creature alive has that and yet, because of those very gifts, we can get stuck in our own heads on big decisions before we make them or sometimes wake up one day to realize we took the wrong path after we did. Building your best life requires making Shut Up. Suit Up. Show Up. a daily practice, just as Hollis suggests. But we have to look inward, too, to assess if we are acting in alignment with our values and our desires. And to know, with confidence, what those values and desires are, and that they are worthy of our time and attention. There's a clear way to do that. Here's how that is done:

Take time every day to update your internal maps.

The other piece of the puzzle is to make sure you know yourself well. Your fears, hopes, aspirations. This kind of

self-knowledge is hard to come by. The combination of life's demands and the ever-present noise of modern life can make it hard to simply know ourselves.

That's where the second part of this protocol comes in. Is this necessary? Yes. People who excessively focus on excelling in outward roles often feel unfulfilled or they end up neglecting other important areas of life. Yet those who are excessively focused on self-knowledge and self-care—another admittedly squishy term—often put off taking action that would allow their careers and relationships to thrive. There needs to be a balance.

The solution, Hollis explains, is to incorporate a second daily practice of exiting the stimulus-response mode that our outward roles—employee, parent, spouse, et cetera—require of us, giving us a chance to direct our attention inward instead. This protocol is like leaving the highway to take a local road: You benefit from the scenery, absence of high-speed traffic, and the insights that inevitably come from this change of pace. This pause allows us to separate what we think other people think about us from what we actually know to be true. That separation improves our awareness of what we want and clarifies what might make our lives genuinely more meaningful and more aligned with who we are.

There are two surefire ways to go about exiting the stimulus-response mode. First, engage in an activity unrelated to any ambitions or outward roles. Draw, read, think, stare at the sky… whatever it is, take time each day for this activity. The key is that the activity doesn't involve any other people, or their expectations or projections of you, or even your dog or cat. That means social media is out; so is posting photos from this activity. This is about you and you alone. Even just ten minutes can make a meaningful difference.

Second, simply close your eyes and ask how you're feeling, and watch for what brings you joy or makes you uncomfortable.

Try one, or both. Each is a way to better know who you are as an individual, and better understand what distinguishes you from others.

Swiss psychiatrist Carl Jung believed that this process is a core aspect of human existence.[59] Hollis's writings and lectures, too, often focus on the importance of this process as a foundation for any other relationship we may have.

These insights might sound wishy-washy or even self-indulgent, but they are neither. Pausing from the

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[59] C. G. Jung, Letters Vol 1, 1906–1950, ed. Gerhard Adler (Routledge, 1973), https://jungiancenter.org/wp-content/uploads/2023/09/letters-of-c-g-jung-vol-1-1906-1950.pdf.

constant need to respond to this or that stimulus gives us the chance to hear, then question, what we have internalized from others, and to hear, then acknowledge, what we know to be true for us, irrespective of what others say. To not explore and continually seek to know our true selves in this way, Hollis explains, is to not live life fully. In his clinical experience, not periodically exiting the stimulus-response mode is what causes people to make choices that can seriously harm themselves or others in the long run.

A different perspective on the benefit of this protocol comes from psychologist Alfred Adler, who believed the need to spend time alone is crucial for us to avoid an inferiority complex, a term he coined.[60] As Adler saw it, the feeling that we aren't as good as other people is something that gets internalized as we grow up. Maybe someone made fun of us for a particular piece of clothing we wore, maybe a parent compared us to an older brother or sister… we all encounter things like this. They are part of life. The problem is that we tend to take them on board. When someone teases us for how we look, we may end up thinking we look bad. When someone expresses disappointment in a grade, we may end up thinking we're not smart. You get the idea. The result is a whole lot of negative opinions about ourselves that we did

[60] Mary C. McCluskey, "Revitalizing Alfred Adler: An Echo for Equality," Clinical Social Work Journal 50, no. 4 (2022): 387–99, https://doi.org/10.1007/s10615-021-00793-0.

not create but that we treat as our own thoughts. Adler saw self-encouragement as a powerful way out of this dynamic.[61]

But how? What would impactful self-encouragement even look like if—as we discussed several times already in chapters 3 and 6—positive self-talk isn't effective?

The answer is: Get yourself out of stimulus-response mode. Not forever. We have to fulfill our roles after all. Just for brief periods of time, but daily. Take a few minutes alone without inputs from others or even your phone.

Spending time alone can help quiet negative voices we've internalized so that we can better hear our own inner voice. It's a skill to learn to hear your own inner voice, and understand it. But it is one well worth cultivating.

This isn't to suggest that we already have the answers to every challenge. Sometimes we need to ask for help, and encouragement, from others. Psychotherapy, for instance, can be extremely beneficial. Data show that it can improve symptoms of psychological difficulties, that these improvements remain stable for years, and that these patients use less healthcare than the population at large after completing therapy.[62]

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[61] Rommy Cedeno and Tyler J. Torrico, "Adlerian Therapy," StatPearls, last updated January 11, 2024, https://www.ncbi.nlm.nih.gov/books/NBK599518/.

[62] Christian Roesler, "Evidence for the Effectiveness of Jungian Psychotherapy: A Review of Empirical Studies," Behavioral Sciences 3, no. 4 (2013): 562–75, https://doi.org/10.3390/bs3040562.

Still, for all these reasons, alone time tends to cause peaceful feelings, even for extroverts. Researchers from the Netherlands and the UK asked 178 people to keep a daily diary for twenty-one days in which they noted how much time they spent in solitude each day, regardless of the activity they were doing, among other reflections.[63] On average, days with more hours in solitude were lonelier for the participants. However, the researchers found that choice made the difference. People were happier being alone when that was what they chose. And across the entire study group, days with more alone time were less stressful. People tended to feel more themselves and more free from external pressure on days when they had comparably more solitude.

Combining alone time with activities that aren't part of the daily demands of life (work, errands, and such) adds to the benefit, because these "extra" activities also tend to be good for our mental health. In September 2023, Nature Medicine published a remarkable study by an international group of researchers in which they looked at data from five long-term studies conducted in sixteen countries enrolling more than 93,000 people.[64] This analysis found that having a hobby

[63] Netta Weinstein et al., "Balance Between Solitude and Socializing: Everyday Solitude Time Both Benefits and Harms Well-Being," Scientific Reports 13, article no. 21160 (2023), https://doi.org/10.1038/s41598-023-44507-7.

[64] Hei Wan Mak et al., "Hobby Engagement and Mental Wellbeing Among People Aged 65 Years and Older in 16 Countries," Nature Medicine 29 (2023): 2233–40, https://doi.org/10.1038/s41591-023-02506-1.

was tied to fewer depressive symptoms, greater happiness, and a greater sense of life satisfaction. The more time spent with these pursuits, the greater these benefits. This study focused on older adults, so one might wonder whether the benefits of a hobby are because they give people who've left the workforce something to do. But the percentage of retirees ranged from 56 percent to 99 percent among the studies included in the analysis, which means that some of the participants still held a job; the benefits cannot be explained by the notion that the hobby is simply replacing a career. Carving out time to pursue our personal interests is good for our health, whether we are still in the workforce or not.

We need time to get to know ourselves. Constantly reacting to external demands leaves no space for thoughts and feelings to surface into our full awareness. It can be tempting to think that who we are is what we do, who we know, where we live—that these are the ingredients of our identity. But when we exit the stimulus-response mode, we can tune in to questions, or even answers, that have been quietly brewing but not fully acknowledged. Hollis speaks about these pauses as moments to "recollect," as in re-collect ourselves. I would encourage you to try it.

If you find it hard to just… sit there in silence, then try a different entry point. I often turn to drawing. There is no

outside demand about what I draw or how I draw it. For you, it may be playing an instrument or gardening. “But those activities have to be done correctly, and I can’t introspect while doing them,” you might say. I get it. There is a format to many such activities, and there are often metrics by which to gauge your success at them. But that doesn’t matter. The key thing is that what you’re doing is not about occupying an outward role that others place on you; it’s about just being you.

It is easy to fall into the trap of thinking we don’t have time for these things. We do. There is always time, if not every day then at least a few times each week. I have an immensely demanding schedule and have for the last thirty or so years. I try to step away from my devices and stand or sit quietly for at least five minutes per day. Time with hobbies can be admittedly more sporadic, based on my work schedule, but I aim to spend at least an hour a week doing something completely unrelated to work or the routine demands of life.

By stepping away, you’ll develop a closer relationship with your own inner voice. Insights and ideas inevitably surface. Not always right away, but everyone I know who does this protocol has described a moment when something related to a thought they had during solo time days prior suddenly shot to the surface of their awareness. The idea was a clear “go,” a call to action. Or a clear “no go,” a call to halt certain

patterns in their life going forward—either small or large. Somehow what was previously suppressed or hidden from their awareness became clear. And it changed their life for the better.

Stories like that are not uncommon, and become the norm when we take the time to regularly exit the stimulus-response mode. They speak to the incredible power of our unconscious mind, but also to our brain as a whole. Most of us have barely tapped into our capacity to link our outward roles and inner narratives, to balance comfort and striving, and to build a life that is uniquely ours. Hollis has said, “Ask large questions, you’ll live a large and interesting journey.”[65] To be clear, I don’t think we are all meant to be artists, just as I don’t think we are all meant to be engineers. I know for certain, however, that we all have an inner and an outward life, and that learning to understand and bridge the two is key to our sense of fulfillment. That, as I understand it, is what Hollis means when he talks about creating our best possible life.

[65] Huberman, “Dr. James Hollis: How to Find Your True Purpose.”

COMMON QUESTIONS

When should I pause for gratitude? In the morning, before I go to work? At night, before I go to bed

There is no time of day that is best for a gratitude practice. But it may work best if you decide on a specific time and place to do so, as a way to boost that conscious awareness and to help yourself stay committed.

Would going for a walk work as a way to exit the stimulus response

Absolutely, as long as you are not turning the walk into a physically demanding exercise (see Exercise Protocol 1 for more about walking, and Exercise Protocol 3 for more about how to know when movement tips over into cardio work). And leave the phone at home. The point here is to give yourself some aimless time where the only variable worth measuring is your connection to yourself.

3

PERSONAL GROWTH PROTOCOL

Resolve Inner Pain Points Through Expressive Writing

WHAT TO DO

- Write for fifteen to twenty minutes about a challenging experience that affected your life. Do this exercise four times, once per day for four consecutive days. Keep in

mind: Do not write for longer than twenty minutes. Set a timer and stop even if you feel you could keep going.

- Do not worry about the quality of your writing. No one will read this but you.
- Write about what happened, what you felt at the time, what you feel now, and any lasting repercussions, however large or small, from this experience. Be honest about everything you felt, and everything you still feel.

HOW IT WORKS

Not everyone has experienced trauma, but if you have, I encourage you to seek out the resources to address it. Whether it happened in childhood or adulthood, trauma and the shadow it leaves can be utterly destructive to our relationship to ourselves, and ultimately to those around us. It can lead to self- or outwardly destructive patterns.[66]

How do we know if we have trauma? Trauma has a specific definition. As Paul Conti, MD, an expert on trauma, explains, it's a negative event, or set of events, that happens to you or that you observe that causes changes in your brain that significantly reduce your functionality in other areas of

[66] Vincent J. Felitti et al., "Relationship of Childhood Abuse and Household Dysfunction to Many of the Leading Causes of Death in Adults," American Journal of Preventive Medicine 14, no. 4 (May 1998): 245–58, https://doi.org/10.1016/S0749-3797(98)00017-8.

life.[67] Symptoms of this can show up in our behavior as hypervigilance, dissociation, or both. It can manifest as sleep disruptions, irritability, or even OCD-like or ADHD-like symptoms.[68] It can also lead to substance abuse. The challenge is that all the symptoms of trauma can arise from other causes as well. There is no blood test or even brain image to rule in or rule out psychological trauma. Not yet anyway.

For now, the only way to know if you have trauma is to seek out a formal assessment from a qualified psychologist or psychiatrist, and as Dr. Conti pointed out, "Always engage the best possible self-care, in or out of treatment." I pressed him a bit more on this.

"As people seek out help, is there anything they can do on their own?" I asked.

Turns out there is. And at first Dr. Conti's answer surprised me. Until I looked at the peer-reviewed data on it. The protocol? Expressive writing. But not a journal entry, or a poem, or a novel. Rather, a very specific form of repetitive recall of a troubling event or events. This is the protocol I mentioned in chapter 6, as a way to undo the kind of rewiring that happens in their wake.

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[67] Paul Conti, Trauma: The Invisible Epidemic (St. Martin's Essentials, 2021).

[68] Kathryn L. Humphreys et al., "Stressful Life Events, ADHD Symptoms, and Brain Structure in Early Adolescence," Journal of Abnormal Child Psychology 47, no. 3 (May 2018): 421–32, https://doi.org/10.1007/s10802-018-0443-5.

Social psychologist James Pennebaker developed the research for this protocol, beginning with a study in the 1980s.[69] He and a colleague asked forty-six participants, all university students, to write about traumatic life events once a day for four days consecutively. For comparison, another group followed the same writing schedule but focused only on trivial occurrences. The researchers collected health records and recorded moods and physical symptoms self-reported by the participants. People writing about traumatic events experienced a decline in mood and an increase in blood pressure immediately following the exercise. But over the following six months, they had fewer visits to the university health center.

Two years later, Pennebaker followed up with another study in which fifty students did the same exercise: Write about a traumatic experience or trivial topic four days in a row.[70] This time, though, the researchers did not use self-reported data. Instead, alongside measuring blood pressure and heart rates before and after the writing exercise, the researchers also collected blood samples and tested blood for two markers of immune system function. Here, again, the

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[69] J. W. Pennebaker and S. K. Beall, "Confronting a Traumatic Event: Toward an Understanding of Inhibition and Disease," Journal of Abnormal Psychology 95, no. 3 (1986): 274–81, https://doi.org/10.1037/0021-843X.95.3.274.

[70] J. W. Pennebaker et al., "Disclosure of Traumas and Immune Function: Health Implications for Psychotherapy," Journal of Consulting and Clinical Psychology 56, no. 2 (1988): 239–45, https://doi.org/10.1037/0022-006X.56.2.239.

exercise proved beneficial to health and mood. Writing about trauma, the researchers concluded, boosted immune function, led to less physical illness, and reduced stress.

A deluge of work on journaling about difficult experiences followed. More than 1,400 papers on expressive writing are now part of the scientific literature, with interest increasing in the forty years since Pennebaker's first studies.[71] A 2004 meta-analysis of nine papers confirmed the improvements in health tied to this type of expressive writing.[72] Among the groups for which this approach has shown a benefit are adolescents, people undergoing cancer treatment, caregivers, people suffering from post-traumatic stress disorder, people experiencing depression, along with many others. Expressive writing isn't a cure. But decades of data provide undeniable evidence that it helps with mental health—and physical health.

Why? What is going on under the surface that makes this simple act work? The mechanisms behind this are multiple, but studies suggest that the most powerful lies in how this type of writing affects the prefrontal cortex. Creating

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[71] Xiaojuan Gao, "Research on Expressive Writing in Psychology: A Forty-Year Bibliometric Analysis and Visualization of Current Status and Research Trends," Frontiers in Psychology 13 (October 2022): 825626, https://doi.org/10.3389/fpsyg.2022.825626.

[72] Pasquale G. Frisina et al., "A Meta-Analysis of the Effects of Written Emotional Disclosure on the Health Outcomes of Clinical Populations," Journal of Nervous and Mental Disease 192, no. 9 (September 2004): 629–34, https://doi.org/10.1097/01.nmd.0000138317.30764.63.

cohesive narratives that are honest—that is, truthful, even when the experience is painful to recount—appears to stimulate this region of the brain in a way that benefits us both psychologically and physically.

Let's zero in on the "honest" part of the equation. A study from 2017 shows the hardwiring of this trait to the prefrontal cortex. Researchers from the University of Zurich, the University of Chicago, and Harvard University gave the 145 study participants a cash reward every time they won a die-rolling game.[73] Only the person who threw the die knew their own results. The test was rigged in a way that put the chances of winning any round at 50 percent.

Here's where the experiment gets interesting: During the challenge, the researchers applied transcranial stimulation (a noninvasive application of a weak electric current to the scalp) to the prefrontal cortex of half the participants. Specifically, this electric current reached the dorsolateral region of the prefrontal cortex, an area activated when we are choosing whether or not to be honest. The group with this stimulation reported winning 58 percent of their rolls, which meant they had lied approximately 15 percent of the time. The group without this stimulation reported winning

[73] Michel André Maréchal et al., "Increasing Honesty in Humans with Noninvasive Brain Stimulation," Proceedings of the National Academy of Sciences U.S.A. 114, no. 17 (April 2017): 4360–64, https://doi.org/10.1073/pnas.1614912114.

68 percent of the time, which meant they had lied about their rolls about 37 percent of the time. The full study included more tests to gain insight into what exactly was going on here. The researchers concluded that this brain region regulates honesty specifically when lying would benefit us.

The amygdala, by contrast, is connected to stress, fear, and anxiety. People under duress tend to have heightened activity in this part of the brain.[74] Activation of the amygdala triggers the release of adrenaline and cortisol,[75] which are crucial for responding to challenges and processing the memory of them. That helps when we're faced with similar challenges in the future. But the overabundance of these chemicals appears to affect the immune system, decreasing our ability to mount an immune response over time.[76]

The specific journaling exercise pioneered by Pennebaker likely works because it avoids activating the amygdala and instead connects to the prefrontal cortex. An added benefit is that any time spent with tasks that deliberately engage the prefrontal cortex may improve brain plasticity, which has a

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[74] Kerry J. Ressler, "Amygdala Activity, Fear, and Anxiety: Modulation by Stress," Biological Psychiatry 67, no. 12 (2010): 1117–19, https://doi.org/10.1016/j.biopsych.2010.04.027.

[75] Benno Roozendaal et al., "Adrenal Stress Hormones, Amygdala Activation, and Memory for Emotionally Arousing Experiences," Progress in Brain Research 167 (2007): 79–97, https://doi.org/10.1016/S0079-6123(07)67006-X.

[76] Jennifer N. Morey et al., "Current Directions in Stress and Human Immune Function," Current Opinion in Psychology 5 (October 2015): 13–17, https://doi.org/10.1016/j.copsyc.2015.03.007.

long-term benefit on cognitive function (for more on this link, turn back to chapter 6).[77]

This type of journaling is not necessarily comfortable. It asks us to revisit the most difficult experiences in our lives—four times. But the point of this exercise is not to feel good in the short term. The point is to create a better relationship with our past, with our emotions about the past, and with ourselves. And remember, this isn't about moving through current conflicts or other difficulties but rather about moving past old wounds that hold us back.

As for doing the writing exercise four times, for four days in a row, the reason is simple. That time frame allows us greater depth of reflection. It takes time to process old wounds, partly because they are just that—wounds. Sticking with this exercise for four days gives you the chance to see how you felt about it then and how you feel about it now, but without that internal dialogue becoming an open-ended rumination, which ends up occupying even more mental real estate. The idea here is to deal with those past pains in a way that lets you make peace with them, and then move forward. Four days is a low commitment. And Pennebaker himself

[77] Hua Tang et al., "Prefrontal Cortical Plasticity During Learning of Cognitive Tasks," Nature Communications 13, article no. 90 (2022), https://doi.org/10.1038/s41467-021-27695-6.

has given this advice: If it doesn't work for you, then stop doing it.[78]

COMMON QUESTIONS

Does it matter what time of day this exercise is done

No. There is nothing about the mechanisms behind expressive writing that draws on physiological changes associated with day and night. Do this protocol at whatever time of day you're able to set aside the time, about fifteen to twenty minutes, plus perhaps a few minutes to recover before moving to the next part of your day.

I did this as described here and I don't feel any better about the past.

This exercise is not designed to resolve emotional pain entirely. It has physical and psychological health benefits, but it is not a cure. Therapy or other interventions are often needed in order to work through emotional pain so that it no longer affects us negatively. Meditation, EMDR, hypnosis, exercise, and other techniques are all widely available. I cannot say what approach will work for you but if you are suffering from emotional pain, then do not hesitate to seek professional help.

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[78] Hayley Phelan, "What's All This About Journaling?," New York Times, October 25, 2018, https://www.nytimes.com/2018/10/25/style/journaling-benefits.html.



PERSONAL GROWTH PROTOCOL

Reframe Loss to Effectively Move Through Grief

WHAT TO DO

- Understand grief as a lack of reward. Over time, your brain has learned to predict certain events, like our expectation of someone's voice or smell. It is a process that draws on our expectation of a "reward" that is no longer there.
- Spend ten to thirty minutes a day to deeply feel your attachment to the person, animal, or thing you are grieving. Do not try to reduce the intensity of your reaction to this attachment. Experience it fully.
- Avoid what-if thoughts. Try not to dwell on how else a situation might have gone.

HOW IT WORKS

Losing people, animals, and things is part of life. It's healthy and normal to experience grief in the wake of those events, provided it eventually wanes. What's "eventually"? That depends. The loss of a parent or sibling is devastating. The loss of a keepsake can be, too, but in a very different way. These are obviously not the same. But no matter who or what we lose, grief has some hallmark features because in every case it is a fracture in our reward-prediction brain

circuitry. What's more, it requires our brain to remap; these circuits, which once had us expecting someone we love to walk in the room, are still there, even if that someone is no longer there. We must reorganize our awareness of what's now in our proximity and come to learn what actions related to the person or thing lost are no longer, well, actionable.

The purpose of this protocol is not to intellectualize, avoid, nor exacerbate grief. It's to ensure you don't get stuck in what is called maladaptive grief, also called complicated or prolonged grief disorder, in which grief interferes with life long after the loss initially occurred, which can be extremely detrimental to your physical and mental health.[79]

I want to note too that this protocol should not replace tools and insights derived from extensive studies of bereavement, like cognitive behavioral therapy for prolonged grief,[80] which call for being wary about heavy drinking in the year or two immediately following a death,[81] and for doing things that keep you in the flow of life like still holding dinner

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[79] Donna M. Wilson et al., "What Exactly Is 'Complicated' Grief? A Scoping Research Literature Review to Understand Its Risk Factors and Prevalence," OMEGA 86, no. 2 (December 1, 2020): 471–87, https://doi.org/10.1177/0030222820977305.

[80] K. B. Komischke-Konnerup et al., "Grief-Focused Cognitive Behavioral Therapies for Prolonged Grief Symptoms: A Systematic Review and Meta-Analysis," Journal of Consulting and Clinical Psychology 92, no. 4 (2024): 236–48, https://doi.org/10.1037/ccp0000884.

[81] János Pilling et al., "Alcohol Use in the First Three Years of Bereavement: A National Representative Survey," Substance Abuse Treatment, Prevention, and Policy 7, article no. 3 (2012), https://doi.org/10.1186/1747-597X-7-3.

plans and getting exercise.[82] Rather, it offers an understanding about grief, based in neuroscience, that may help you cope with these strong and difficult feelings that come in after loss.

The first component of this framework requires a shift in thinking: Consider grief not as a state of sadness but rather as a state of longing. Brain-imaging studies have found that grief activates regions and circuitries that are involved with motivation and craving. A study led by neuroscientist and grief researcher Mary-Frances O'Connor, at UCLA, proved just that. O'Connor and her team imaged the brains of twenty-three women who were actively experiencing grief, eleven of whom were experiencing a state referred to as prolonged or complicated grief,[83] generally defined as intense, debilitating, and unrelieved grief lasting more than a year. The women viewed photographs of the deceased and words related to their death as brain images were taken using functional magnetic resonance imaging.

Brain images of the women with complicated grief showed activity in pathways connected to motivation and reward during moments when the women reported feelings of

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[82] "Coping with Grief and Loss," National Institute on Aging, content reviewed July 15, 2024, https://www.nia.nih.gov/health/grief-and-mourning/coping-grief-and-loss.

[83] Mary-Frances O'Connor et al., "Craving Love? Enduring Grief Activates Brain's Reward Center," NeuroImage 42, no. 2 (2008): 969–72, https://doi.org/10.1016/j.neuroimage.2008.04.256.

yearning. These neural circuitries lead to the release of chemicals that make us feel good, like dopamine, serotonin, and oxytocin, keeping us motivated to obtain things that are good for us, like food and social connections. Specifically, the images showed activation of the nucleus accumbens, which releases dopamine during social interactions.[84] This (at the time) surprising observation led the researchers to hypothesize that grief is akin to searching for a reward that never arrives. We stay motivated to keep finding it, but we don't—and can't. That makes sense to anyone who has experienced grief but until O'Connor's work, neuroscientists were unaware that grief was tied to reward anticipation circuits in the brain.

We all experience loss. This is a lesser example than the one described above, but simply for example: Anyone who's lost a large bill knows what it's like to look in their wallet and not find it. You can tell yourself it's just money, and you will get over it, but the loss of it brings the sudden understanding that whatever it made possible is now gone. Translate that to more emotional losses, such as a loved one or a friend or a pet. Suddenly our neural circuitry around that attachment is no longer actionable. The pathways that used to fire are still there, but they aren't useful anymore. What is possible has

[84] Anne F. Pierce et al., "Nucleus Accumbens Dopamine Release Reflects the Selective Nature of Pair Bonds," Current Biology 34, no. 3 (2024): 519–30.e5, https://doi.org/10.1016/j.cub.2023.12.041.

changed. The circuit of anticipation and action resolution remains open. And that's painful.

But motivation circuitries aren't the only ones at play in this state of longing. Our brains contain cells dedicated to mapping the world we move through—think of it like a sort of mental map of where things are.[85] Likewise, we also map our relationships, keeping track of the ties that connect us, albeit with different brain cells.

Studies have found some overlap among neurons we use to orient ourselves in space and time and those we use to understand our relationships to other people.[86] Researchers at Dartmouth College found that physical distance, proximity in time, and social closeness activate the same region of the brain, namely the inferior parietal lobe.[87] The circuitries that store information about the distance from

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[85] Edvard I. Moser et al., "Place Cells, Grid Cells, and the Brain's Spatial Representation System," Annual Review of Neuroscience 31 (July 2008): 69–89, https://doi.org/10.1146/annurev.neuro.31.061307.090723; Valeria C. Caruso et al., "Compensating for a Shifting World: Evolving Reference Frames of Visual and Auditory Signals Across Three Multimodal Brain Areas," Journal of Neurophysiology 126, no. 1 (April 2021): 82–94, https://doi.org/10.1152/jn.00385.2020; Russell A. Epstein et al., "The Cognitive Map in Humans: Spatial Navigation and Beyond," Nature Neuroscience 20, no. 11 (November 2017): 1504–13, https://doi.org/10.1038/nn.4656.

[86] Michael Peer et al., "Brain System for Mental Orientation in Space, Time, and Person," Proceedings of the National Academy of Sciences 112, no. 35 (2015): 11072–77, https://doi.org/10.1073/pnas.1504242112.

[87] Carolyn Parkinson et al., "A Common Cortical Metric for Spatial, Temporal, and Social Distance," Journal of Neuroscience 34, no. 5 (2014): 1979–87, https://doi.org/10.1523/JNEUROSCI.2159-13.2014.

your bed to the bathroom, from your house to the supermarket, and from your town to Paris overlap with circuitries that store information about how close another person is to you socially.

Rewiring the neural circuitries that orient us with regard to the people in our lives takes time. We experience this physically often: If you've ever rearranged the furniture in your living room, for example, you know that it can take time to remember that the couch isn't where it used to be. Grieving includes a similar reorientation but, of course, with heightened emotionality. We have to get used to our loved one not being where we always found them—arriving home from work at 6:00 p.m., at the other end of the phone, across from us at the dinner table. We no longer experience the rush of chemicals we once did when that happened, what our brain registered as a reward. Grief exists where these circuitries contradict with reality, what O'Connor refers to as the "Gone-But-Also-Everlasting" theory of grief.[88] It takes time for our brains to update. Thus grief is not only a disruption to our circuitries connected to motivation and reward, but also a difficult learning process.

[88] Mary-Frances O'Connor and Saren H. Seeley, "Grieving as a Form of Learning: Insights from Neuroscience Applied to Grief and Loss," Current Opinion in Psychology 43 (February 2022): 317–22, https://doi.org/10.1016/j.copsyc.2021.08.019.

Avoidance makes this learning process even harder because it prevents us from incorporating new information into the map.[89] If we are refusing to face reality, then our brains can't integrate it. As O'Connor has written, "Learning requires experience, and avoidance of things that remind the bereaved of the loved one's death or absence will prevent accommodation of the information."[90]

Instead, allowing ourselves to really feel our attachment to a person we've lost, a pet who's died, or even a missing object that held sentimental value, without denying what happened, gives our brains a chance to decouple the object of our love from our current environment and our brain's expectation of being able to interact with the person (or pet) that was lost. This decoupling—accepting a death while also savoring the memories and the essence (meaning our internal representation of the person themselves)—allows us to hold on to these important feelings without triggering a prediction that, say, the person will be waiting for us in the kitchen when we arrive home from work. Taking time to feel this attachment allows our brains to consolidate new memories and create maps, or expectations, about our lives.

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[89] Amanda W. Baker et al., "The Role of Avoidance in Complicated Grief: A Detailed Examination of the Grief-Related Avoidance Questionnaire (GRAQ) in a Large Sample of Individuals with Complicated Grief," Journal of Loss and Trauma 21, no. 6 (February 2016): 533–47, https://doi.org/10.1080/15325024.2016.1157412.

[90] O'Connor and Seeley, "Grieving as a Form of Learning."

Some researchers suggest what they call grief rituals as a way to "be" with a deceased loved one.[91] A person might light a candle, spend time with a photograph, or write a letter to the person, for example. Visiting a grave can be understood as a grief ritual. These ceremonial acts allow us to experience feelings of loss in a way that helps us begin to separate the person (or other object of our grief) from our new reality.

Finally, a word about counterfactual thinking that sometimes arises during the grief process, those "what-if" thoughts that have us imagining a different outcome if only we'd come home earlier, taken a different route, said a kind word, or spent more time with them. A psychologist friend of mine once told me "if only" are the two most destructive words we can think, say, or write. Had I known that long ago, I would have saved myself a lot of pain, and had I known the modern neuroscience of grief, I would have understood that pain I felt (and sometimes still feel) is one of yearning, a natural reflex to act—call, write, see that person—that is impossible to carry out.

It's a mistake to wonder why we still aren't over that pain, to suppress it. But the biggest mistake is to ruminate in "if

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[91] Joanna Wojtkowiak et al., "Ritual in Therapy for Prolonged Grief: A Scoping Review of Ritual Elements in Evidence-Informed Grief Interventions," Frontiers in Psychiatry 11 (February 2021): 623835, https://doi.org/10.3389/fpsyt.2020.623835.

only.” That type of thinking inhibits the grief process. It’s a lie about an alternate reality that truly blocks recovery. In one study, researchers examined links between counterfactual thoughts and symptoms stemming from prolonged grief and depression among fifty-nine people suffering a recent loss.[92] They found a strong correlation. People who engaged in what-if thinking experienced a longer period of grieving and symptoms of depression.

Besides, grief is hard enough without the guilt. Psychology, philosophy, religion—all these disciplines have considered grief, tried to help us understand why we experience it and how to live with it. The protocol above is one of neuroscience’s offerings. Understand that grief is like a craving for a reward that is no longer coming. Make space to feel the deep attachment, whether through small rituals or just finding moments in your day to pause and fully experience your feelings. This time will allow your brain circuitries to rewire, creating memories along with new mental maps. Finally, resist engaging in counterfactual thinking, which can prolong suffering, trigger feelings of guilt, and exacerbate any depressive symptoms arising during this period.

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[92] Maarten C. Eisma et al., “Upward and Downward Counterfactual Thought After Loss: A Multiwave Controlled Longitudinal Study,” Behavioral Therapy 52, no. 3 (2021): 577–93, https://doi.org/10.1016/j.beth.2020.07.007.

COMMON QUESTIONS

What is a healthy amount of time for grief

Psychologists tend to consider prolonged grief to be grief that is debilitating and lasts more than a year. With prolonged grief, people feel depressed and hopeless, often stop engaging socially, and may have little appetite.

Not all grief is like this, but it’s hard to put a time frame on when any grief should end, since the process is so personal. It’s more important to understand what healthy grieving is, rather than putting a time frame on the process. Allow yourself the time and space to be with your deep feelings of attachment and closeness and let the process unfold

What I can say, though, is that one day, you might be walking along and remember someone you lost, and you might experience that person—not just the memory of them, but them—as a completely positive presence that somehow (and I don’t have any science to explain this) is fully integrated into who and where you are now, right at that moment. Meaning: You know they are gone, and yet they are still very present. The remapping has occurred. That’s happened in my life many times over. To me, that is the end stage of the grief process, one that fortunately never subsides. It just takes some time to get there.